

IMO 2026 — Problem 4

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

partial

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Approaches tried

- Initial analysis of one-move win conditions. Realized Mulan needs both resulting triangles to be winning.

- Quick discretized search for some α values (depth 3-4, limited split points). Results:
 - 45° , 60° , 90° appear winning (no counterexample found).
 - 30° mostly winning but $(91, 44, 45)$ loses.
 - 36° appears winning.
 - 72° , 108° , 120° , 150° lose from many states.
- Formalized the operation algebraically: split angle a into p , q ; triangle options are $(p, b, c+q)$ and $(q, c, b+p)$.
- One-move win analysis: Mulan can win in one move if she can split such that both resulting triangles contain α .
 - This is possible when $\alpha = 90^\circ$ (if both other angles $< 90^\circ$, split as $90 - b$ and $90 - c$).
 - Also possible if some angle $= k$? But need both triangles to have α .
- Search too slow to explore thoroughly. Need a theoretical characterization.

Current best

We suspect the answer might be: Mulan can force a win if and only if α is not a divisor of 180° in the sense that $180^\circ/\alpha$ is not an integer? Or perhaps $\alpha = 90^\circ$ is the only winning? But search suggests 36° , 45° , 60° also winning. Let's compute more precisely.

The state space is continuous; discretization may miss moves. Need a mathematical invariant or strategy.

Full proof

(Not yet complete)